



Pass 1 — Full window (Phase-A only)

Free: $t_0, I_{ss}, I_{sub}, \tau_{ac}, \tau_{dc}, \phi_a$ (6 params)

Fixed: none

Objective: $\min \|\mathbf{r}_A\|^2 = \|i_a^{meas} - i_a^{model}\|^2$

Reliable outputs: $\hat{t}_0, \hat{I}_{sub}, \hat{\tau}_{dc}, \hat{\phi}_a$

Unreliable: $\hat{I}_{ss}, \hat{\tau}_{ac}$ (envelope/SS aliasing)

lock

Pass 2 — Tail window (Phase-A)

Free: I_{ss}, τ_{ac} (2 params)

Fixed: $\hat{t}_0, \hat{I}_{sub}, \hat{\tau}_{dc}, \hat{\phi}_a$

Objective: $\min \|i_a^{meas}[\text{tail}] - i_a^{model}[\text{tail}]\|^2$

Why tail? Transient decayed $\Rightarrow$ envelope $\approx I_{ss}$ — well-conditioned

Merge

$$\hat{\theta} = \{\hat{t}_0, \hat{I}_{sub}, \hat{\tau}_{dc}, \hat{\phi}_a\}_{P1} \cup \{\hat{I}_{ss}, \hat{\tau}_{ac}\}_{P2}$$

Full residual $\mathbf{r}(\hat{\theta})$ recomputed

$\hat{\theta}, \|\mathbf{r}\|, J$ (to confidence scorer)